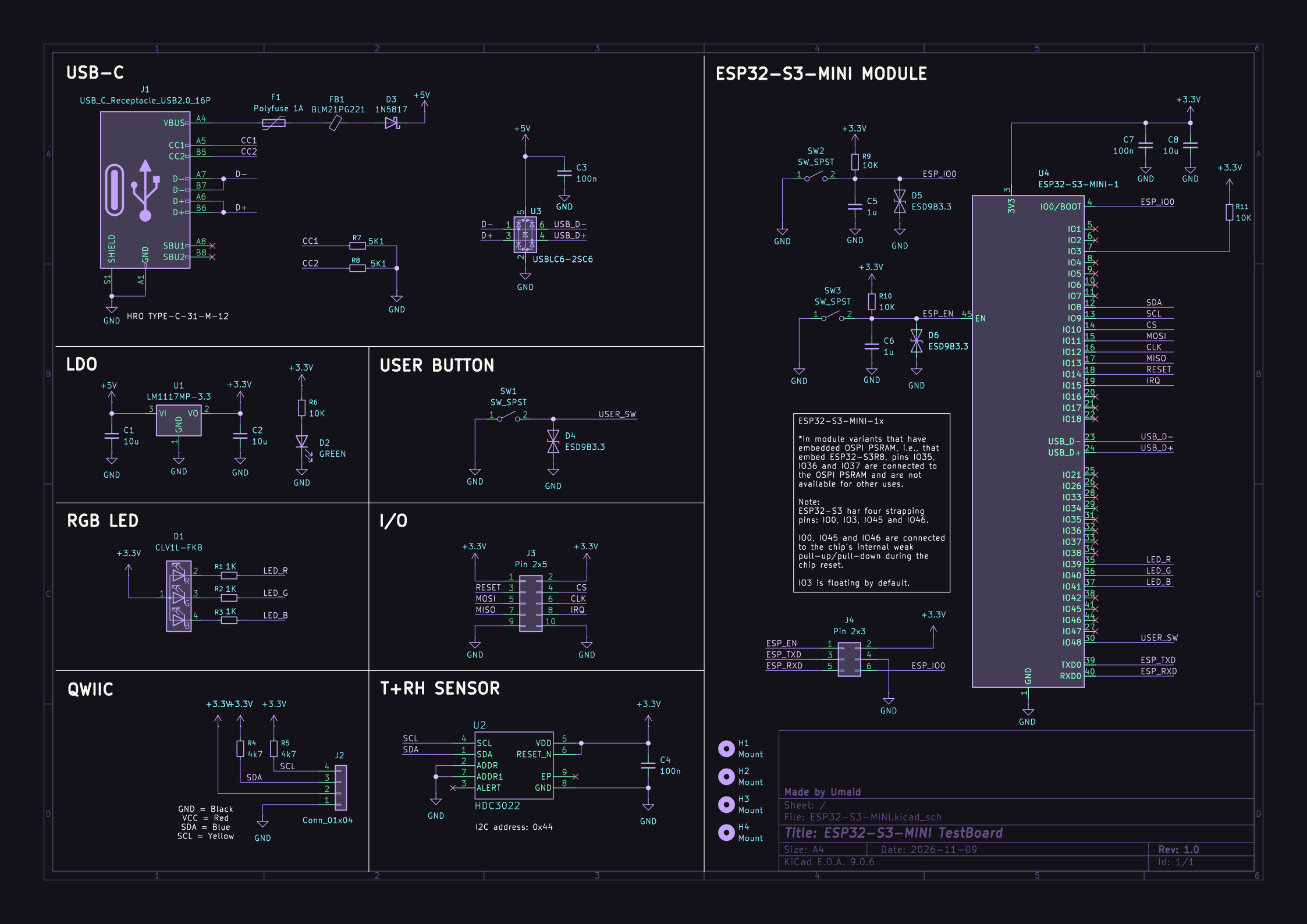

USB-C
J1
USB_C_Receptacle_USB2.0_16P
F1
Polyfuse 1A
FB1
BLM21PG221
D3
1N5817
+5V
VBUS
A4
CC1
A5
CC2
B5
D-
A7
B7
D+
A6
B6
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
HRO TYPE-C-31-M-12
R7 5K1
R8 5K1
U3
USBLC6-2SC6
C3
100n
USB_D-
USB_D+
LDO
U1
LM1117MP-3.3
VI
VO
C1
10u
C2
10u
R6
10K
D2
GREEN
USER BUTTON
SW1
SW_SPST
USER_SW
D4
ESD9B3.3
RGB LED
D1
CLV1L-FKB
R1 1K
R2 1K
R3 1K
LED_R
LED_G
LED_B
I/O
J3
Pin 2x5
RESET
MOSI
MISO
CS
CLK
IRQ
QWIIC
R4
4k7
R5
4k7
SCL
SDA
J2
Conn_01x04
GND = Black
VCC = Red
SDA = Blue
SCL = Yellow
T+RH SENSOR
U2
HDC3022
ADDR
ADDR1
ALERT
RESET_N
EP
C4
100n
I2C address: 0x44
ESP32-S3-MINI MODULE
SW2
SW_SPST
R9
10K
C5
1u
D5
ESD9B3.3
ESP_IO0
SW3
SW_SPST
R10
10K
C6
1u
D6
ESD9B3.3
ESP_EN
U4
ESP32-S3-MINI-1
C7
100n
C8
10u
R11
10K
IO0/BOOT
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO21
IO26
IO33
IO34
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
USB_D-
USB_D+
TXD0
RXD0
3V3
EN
ESP32-S3-MINI-1x
*In module variants that have embedded OSPI PSRAM, i.e., that embed ESP32-S3R8, pins IO35, IO36 and IO37 are connected to the OSPI PSRAM and are not available for other uses.
Note:
ESP32-S3 har four strapping pins: IO0, IO3, IO45 and IO46.
IO0, IO45 and IO46 are connected to the chip's internal weak pull-up/pull-down during the chip reset.
IO3 is floating by default.
J4
Pin 2x3
ESP_TXD
ESP_RXD
H1 Mount
H2 Mount
H3 Mount
H4 Mount
Made by Umaid
Sheet: /
File: ESP32-S3-MINI.kicad_sch
Title: ESP32-S3-MINI TestBoard
Size: A4
Date: 2026-11-09
Rev: 1.0
KiCad E.D.A. 9.0.6
Id: 1/1